

Donald Heddeshheimer

Atlanta, GA | donalddhedges@gmail.com | [linkedin.com/in/donalddhedgesheimer](https://www.linkedin.com/in/donalddhedgesheimer)

EDUCATION

Georgia Institute of Technology Honors Program

Expected May 2028

B.S. in Computer Science – Systems Architecture & Devices | GPA: 3.95/4.00

Coursework: Operating Systems, Processor Design, Computer Architecture, Computer Networks, Algorithms

EXPERIENCE

Solopulse

Peachtree Corners, GA

Software Engineering Intern

May 2026 - Present

- Engineered a real-time radar direction-finding pipeline in **C++/CUDA**, scanning 131,000+ candidate directions per frame across various sensor channels; deployed and validated during live flight trials at White Sands Missile Range.
- Cut per-packet signal-processing latency by $\sim 8\times$ by profiling with **NVTX** and **NVIDIA Nsight Systems**, migrating bottleneck operations to the GPU, and introducing plan caching and event-driven scheduling.
- Built an end-to-end emitter-geolocation system that fused radar bearings with aircraft position and attitude, intersected lines of bearing with WGS-84, and streamed location estimates to ground software over binary TCP.
- Refactored core processing into a testable state machine and built a recorded-sensor replay harness, raising automated test coverage to over 90% across 160 GoogleTest cases.

RoboNav (RoboJackets) – University Rover Challenge

Atlanta, GA

Autonomous Navigation Software Developer

January 2025 - May 2026

- Cut median replans by 45% on a fixed obstacle-course suite by unifying planner/controller grid-map semantics and feeding Gaussian-inflated LiDAR terrain costs into a custom **ROS2 C++ A*** planner on a Jetson Orin Nano.
- Achieved up to 2.5 cm global position accuracy under RTK-fixed conditions using a dual-EKF architecture that fused base station corrected RTK GNSS, wheel odometry, and VectorNav IMU data for smooth local motion and global drift correction.
- Accelerated full Gazebo navigation regression suite by **10 $\times$** using parallel, parameterized scenarios built from competition-site DEMs and calibrated robot/sensor models; gated every merge with builds, tests, linting, and formatting in GitHub CI.

RESEARCH

Laboratory for Intelligent Decision and Autonomous Robots

Georgia Tech | Atlanta, GA

Undergraduate Research Assistant

August 2025 - May 2026

- Co-developed an RL-augmented model predictive control (MPC) framework for bipedal locomotion that achieved **86% success versus a 1% baseline** on 10-cm stairs in NVIDIA IsaacLab by adapting its dynamics model, swing trajectories, and gait timing.
- Built a sparse GPU optimization solver that ran MPC across 4,096 IsaacLab environments in parallel, cutting solve time from **10+ seconds to about 200 ms ($\sim 50\times$)** and accelerating end-to-end policy training by over $20\times$.
- Implemented FastSAC and PPO in the lab's PyTorch-based RLOpt framework and repaired FastSAC's replay-buffer insertion and sampling path, restoring stable multi-hour humanoid imitation training with sustained return and episode-length gains.

Big Data Big Impact @ GT: Agricultural Productivity Forecast for Zero Hunger

Atlanta, GA

Machine Learning Researcher

August 2024 - May 2025

- Worked with a research team to build a three-stage agricultural ML pipeline combining satellite and tabular data, achieving 0.92 Dice for U-Net cropland segmentation, 89% accuracy for 10-class LSTM crop classification, and $R^2 = 0.977$ for yield regression.
- Cut multi-terabyte satellite-data preprocessing time by **up to 100 $\times$** by parallelizing Sentinel Hub API requests and building a Redis geospatial cache that processed only uncached portions of overlapping Landsat and Sentinel tiles.

PROJECTS

cuCadence – <https://github.com/donald-heddeshheimer/cadence>

- Created cuCadence, a header-only C++17/CUDA profiler, to complement my Nsight optimization work at Solopulse with continuous in-process visibility into stage latency, tail spikes, and deadline misses in real-time GPU loops.
- Achieved 3.4 μ s per GPU scope (48% lower overhead than blocking CUDA-event timing) through deferred event resolution, thread-local buffers and event pools, and bounded-memory online statistics.
- Validated cuCadence on llama.cpp's CUDA backend with no measurable throughput loss at graph-level instrumentation, then hardened the library with CUDA Graph capture safeguards and regression fixes for three reporting defects.

SKILLS

Languages: C++17/20, CUDA C++, Python

GPU & Real-Time Systems: NVIDIA Nsight Systems/Compute, NVTX, CUDA Events/Graphs, Perfetto, multithreading, TCP/IP, Linux

Robotics & Control: ROS 2, Gazebo, NVIDIA Isaac Lab, Jetson Orin Nano, MPC, EKF, A*, RTK GNSS, URDF/Xacro

ML & Data: PyTorch, PPO, SAC, U-Net, LSTM, Weights & Biases, Redis

Development & Infrastructure: CMake, GoogleTest, Git, GitHub Actions, Docker, Slurm